



Intentionality matters for third-party punishment but not compensation in trust games

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ABSTRACT

We investigate how the intentionality of investors or trustees' actions affects third party compensation and punishment interventions after a trust game. Investors and trustees are randomly assigned to conditions where they either make intentional choices or their choices are made by a random machine. Overall, we find that lack of reciprocity is punished more than lack of trust, and third parties exhibit strong preferences for compensation over punishment. We find that only the punishment choice is affected by the intentionality of parties' actions, whereas compensation occurs in all conditions, whether lack of trust or reciprocity has been intentional or unintentional.

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1. Introduction

Without legal enforcement, social norms rely on informal sanctions to produce conformity. Empirical studies have reported evidence of sanctions both by parties directly involved in the norm-governed interaction (de Quervain et al., 2004; Nelissen and Zeelenberg, 2009; Seip et al., 2014), and by uninvolved third parties (Fehr and Fischbacher, 2004; Kahneman et al., 1986; Kurzban et al., 2007). Whereas positive sanctions (e.g., rewards) and negative sanctions (e.g., punishment) encourage conformity with social norms, we also often observe compensation of victims of norm violations. Compensation does not directly enforce norms. It rather *honors* norms, as it points to the fact that the victim ought to have received fair treatment. If indeed compensation has an important norm-signaling function, we would expect both regulatory mechanisms, compensation and punishment, to be employed in upholding social norms.

The current literature on reactions to injustice by third parties shows that there are two ways observers might react: they could either punish the perpetrator or compensate the victim (e.g., Adams and Mullen, 2015; Chavez and Bicchieri, 2013; Darley and Pittman, 2003; FeldmanHall et al., 2014; Leliveld et al., 2012; Lotz et al., 2011; Van Doorn et al., 2017, 2018; Van Prooijen, 2010; Van de Vyver and Abrams, 2015). Recent studies including both punishment of a perpetrator and compensation of a victim as behavioral options reported that third parties seem to prefer compensation (Chavez and Bicchieri, 2013; Leliveld et al., 2012; Lotz et al., 2011; Van Doorn et al., 2018), even when compensation is costly. One reason for this preference may lie in the fact that compensation has the direct benefit of repairing the harm done to the victim, whereas punishment of a perpetrator has the uncertain benefit of deterrence (Van Doorn et al., 2018). Another reason may be that punishment may elicit retribution, whereas compensation may be socially rewarded in the form of gaining approval and respect, as well as future reciprocal help (Haesevoets et al., 2014; Hart et al., 2019; Lotz et al., 2011). Third parties that

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compensate victims enjoy reputational benefits compared to parties that punish perpetrators. They tend to be perceived as more trustworthy and are preferred as partners in cooperative interactions that involve trust (Dhaliwal et al., 2021).¹

In all the above studies, punishment or compensation refer to a violation of a social norm of fairness or reciprocity, but usually no independent evidence is presented about the players' consensus about the norm's existence and relevance to the experimental situation. In the definition that we adopt, a social norm (Bicchieri, 2006, p.11) is a behavioral rule for which it must hold for sufficiently many people that: 1. they know that such a behavioral rule exists; 2. they prefer to comply with the rule if they believe a) that others will also comply (empirical expectations) and b) that others believe they ought to comply and might sanction non-compliance (normative expectations). While observed behavior remains a crucial measure of compliance with elicited norms, recent experimental work has introduced the explicit use of surveys to assess normative consensus (Bicchieri and Chavez, 2010; Bicchieri et al., 2020; Bicchieri et al., 2011; Krupka and Weber, 2013; Rauhut and Winter, 2010; Reuben and Riedl, 2013). In this paper, we measure both the expectations and behavior of third parties to assess the presence and strength of social norms that may guide their choices. In our simplified version of the trust game, the investor has the choice of transferring the entire endowment to a trustee or keep it all for themselves. If the transfer occurs, the trustee has the choice of sending back half of the quadrupled endowment to the investor, or keep it all. Though the literature has presented both trusting and reciprocating as social norms (Dawes et al., 1991; Orbell and Dawes, 1991), the first of our pre-registered hypotheses is that reciprocity is a strong social norm, whereas trusting is a weak norm at best, if at all (Bicchieri et al., 2011). To this effect, we elicit normative and empirical expectations of third parties about the behavior and beliefs of other third parties. The data support our hypothesis, as we find our participants hold a strong norm of reciprocity. There is little agreement that lack of trust is blameworthy and should be punished, whereas lack of reciprocity is highly reproachable. It should also be noted that third-party behavior is consistent with their empirical and normative expectations. That is, when a reciprocity norm is violated, punishment will occur much more often and severely than when there is lack of trust.

Our second hypothesis is that third parties will employ both compensation and punishment when these regulatory mechanisms are available. We investigate whether the presence of both mechanisms leads individuals to discount one in favor of the other.² In line with other studies, we find that compensation is preferred to punishment. Unlike other studies, however, we explore whether punishment and compensation equally *signal* the presence of social norms, or instead have unequal signaling power. What we find is that compensation and punishment have unequal signaling power. More specifically, while punishment is a strong signal of the presence of a social norm, compensation is not.

Our third hypothesis is that intentionality will matter when making punishment and compensation decisions.³ To test this hypothesis we introduced a version of the trust game in which either the investor or the trustee made an involuntary choice, and this was common knowledge. In one treatment, the investor's choice is made by a random machine that chose to send money with only a 50% probability; in another treatment, the trustee's choice is made by a random machine that chose to return half with 80% probability, nothing otherwise. In both versions, one of the player's actions is therefore unintentional.

We normally punish intentional transgressions, as we consider the transgressor responsible for their actions. Punishment is generally not meted if the behavior is unintentional. What about compensation? Does intentionality matter when the perpetrator acted without the intent to harm? We show that compensation occurs in all conditions, whether the harm has been intentional or unintentional. Punishment instead regularly occurs in case of norm violations when behavior is intentional. If the behavior is unintentional, we find that there is virtually no punishment. Punishment therefore is a strong signal of norm violation, whereas compensation is only a weak signal, as it occurs anyway. This is not surprising, since we often compensate victims of natural disasters or other adverse events that have nothing to do with norm violations.⁴ Intentionality thus plays a crucial role in determining if punishment should occur, but plays no role in deciding to compensate victims of harm. We also find that compensation does not discriminate between trust and reciprocity violations, and in this sense it is not a strong signal that a norm has been violated. Punishment instead is frequent and substantial only when reciprocity is violated.

Our data also show that while compensation is not a signal of norm violation, compensating victims of intentional or unintentional harm is a social norm itself. Indeed, a majority of third-party observers expect other third parties to compensate harm victims in any circumstance (either for lack of trust or lack of reciprocity) and irrespective of violators' intentions. Majority normative expectations were also in favor of compensating both victims of intentional and unintentional lack of trust and reciprocity. Behavior of third parties was consistent with their empirical and normative expectations, suggesting that a norm of compensation is in place.

In what follows, we describe the experimental design in Section 2. Section 3 reports on the results regarding third-party actions and beliefs across different intentionality conditions. We discuss our key findings in Section 4.

¹ Though recent studies by Jordan et al. (2016) and Jordan and Rand (2020) show that third party punishment, too, can serve as a reputation heuristics,

² For example, the opportunity to compensate victims might focus third parties on feelings of compassion, which has been found to decrease punitive sentiment (Condon & DeSteno, 2011).

³ In our legal system, intentionality is a crucial discriminant between crime and misdemeanor. Intentionality is related to responsibility, moral or otherwise, that the law takes into consideration in deciding the level of punishment.

⁴ The motivation to compensate is strongly linked to empathy, whereas it has been shown that anger is a main driver of punishment (Thulin & Bicchieri, 2016).

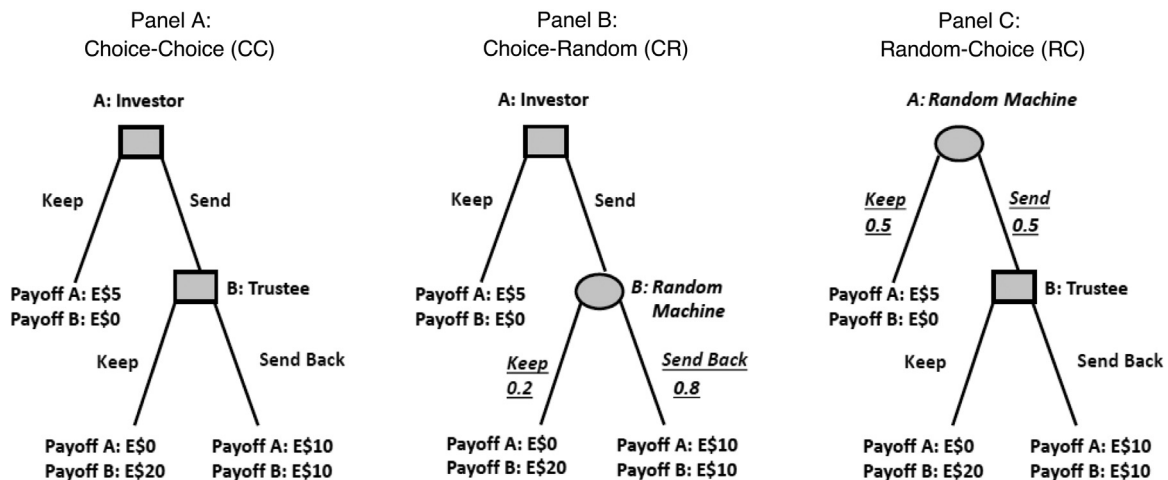


Fig. 1. Experimental Conditions.

2. Experimental design

The key subjects in our experiment are third-party observers. They were presented with the outcome of a previously conducted simplified trust game between an investor and a trustee (Berg et al., 1995). The investor received an endowment of 5 experimental dollars (E\$5) and had to decide whether to keep the endowment or transfer it to the trustee. If the investor decided to keep the endowment, the game ended with the investor's endowment at E\$5 and the trustee's endowment at E\$0. If transferred, the endowment was quadrupled by the experimenters and the trustee received E\$20. The trustee then had to decide between keeping the entire enlarged endowment and leaving the investor with E\$0 or transferring back half to the investor, thus having them both end the game with E\$10. Four experimental dollars (E\$4) equaled one USD dollar (\$1). As we explain in the next subsection (2.1), we varied the intentionality of trust game players' decisions, i.e. the choices of sending/keeping the endowment were made by the players or by a random machine.

The third-party observers were endowed with E\$10, and were randomly assigned to one of the conditions in a 2×3 between-subjects design that included two outcome conditions (lack of trust/reciprocity) and three intentionality conditions (intentional/unintentional choice by investors/trustees). Third parties could choose to add and/or deduct from the final payoffs of investors and/or trustees or do nothing. Neutral framing was used for all subjects (Players A, B, C) and their actions (sending, keeping, adding, or deducting from the endowment).

A total of 128 subjects participated as investors and trustees in the experiment that lasted approximately 20 min. They earned a \$2.4 participation fee and a median amount of \$2.8 in the trust game after observers' interventions⁵ (their average earnings before the interventions were \$1.88).⁶

2.1. Intentionality conditions

A randomly matched pair of an investor and a trustee were randomly assigned to one of the three intentionality conditions (Fig. 1). All of the following intentionality conditions were common knowledge among the players. The first intentionality condition was a choice-choice (CC) condition, where the investor and trustee made their own decisions in the game. The second semi-intentionality condition was a choice-random (CR) condition, where the investor made their own decision in the game, while the trustee's decision was determined by a random machine. The machine had an 80% probability of returning half of the enlarged endowment to the investor and a 20% probability of keeping it, in accordance with prior trust games (Fetichenhauer and Dunning, 2009; Thulin and Bicchieri, 2016). Hence the trustee did not have an active role in this condition.

The third semi-intentionality condition was a random-choice (RC) condition, where the investor's decision was determined by a random machine and the trustee made their own decision. The random machine had a 50% chance of keeping

⁵ Since there were more observers than trust game players, we randomly assigned multiple observers to each trust game player pair and changed their final endowments by the sum of the changes made by all the observers assigned to them. This way, observers' decisions made a real impact on the final payoffs of investors and trustees. Observers were aware that there were other observers in the same experiment, but they did not know that there were other observers assigned to the same investor/trustee pair, to avoid free-riding incentives and diffusion of responsibility. Investors and trustees received their participation fees within 3 days of completing the experiment and the rest of their experimental earnings (adjusted for the observers' interventions, and always positive for each player) within 7 days of completing the experiment.

⁶ The instructions given to investors and trustees stated explicitly that they would directly and indirectly interact with other participants and that their final earnings would depend on the decisions they and others make.

the investor's endowment and a 50% chance of transferring it to the trustee. The percentages are based on the results of prior trust games (Berg et al., 1995; Camerer, 2003; Johnson and Mislin, 2011). The investor did not have an active role in this condition.⁷

2.2. Outcome conditions

The trust game had three possible outcomes: (5,0), (0,20) and (10,10). Across the intentionality conditions, the investor showed an intentional lack of trust towards the trustee if they decided to keep the endowment (CC, CR). In the RC condition, lack of trust was unintentional. The trustee showed lack of reciprocity towards the investor if they decided to keep the entire enlarged endowment (CC, RC). In the CR condition, lack of reciprocity was unintentional. Lastly, the investor showed trust if they sent their endowment and the trustee showed reciprocity if they sent back half of what they received, improving the initial positions of both players.

The third-party observers were endowed with E\$10, and were randomly assigned to one of the conditions in the 2×3 between-subjects design that included two outcome conditions and three intentionality conditions. The third-parties were made aware that they had the opportunity to change the payoffs of real trust game players and were assured their decisions would be implemented. We explore the two outcome conditions where the investor keeps the endowment showing a lack of trust ("No Trust" condition) and where the trustee does not transfer back half of the multiplied endowment to the investor showing a lack of reciprocity ("No Reciprocity" condition). As social norm violations are the focus of this paper, the trust games where the investor sent the endowment and the trustee returned half of it are not included in the analyses (full data are available in Table A5 in the Appendix).

2.3. Belief elicitations and actions of third-party observers

Before the outcome of the trust game was shown to the observers, they were made aware of the specific intentionality condition and players' common knowledge of them. Next, the outcome of the trust game was shown and the observers had the opportunity to compensate and/or punish both, one or none of the trust game players at a cost to their own initial endowment of E\$10. It cost the observer E\$0.25 to compensate or punish a player by E\$1. The only restrictions the observers had in compensating and punishing the players were to not spend more than their E\$10 endowment and to not make any player's endowment negative by punishment. Their decisions were implemented and money was added to or taken out of the investors' and/or trustees' final payoffs.

After making the compensation and punishment decisions, the observers answered questions regarding their empirical expectations (EE) about the actions of other third parties, their own personal normative beliefs (PNB) with respect to adding money to investors (trustees) and deducting money from trustees (investors), and their normative expectations (NE) regarding the normative beliefs of other third parties (see questions in Table 1).⁸ Observers were divided into three distinct groups, depending on the intentionality conditions in the trust game (CC, RC, CR). In each group observers were providing their EE and NE about other observers in their group. The questions referred to the situation in which investors/trustees may have shown lack of trust/lack of reciprocity. The aim of the questionnaire was to test for the existence of social norms via observers' social expectations. The observers were incentivized to give their best estimates from 0% to 100% in increments of 25% and earn \$0.1 per correct answer. Note that normative expectations are second-order beliefs about what other observers approve or disapprove of. Personal normative beliefs questions were included to test for the existence of pluralistic ignorance, a situation where a majority privately accepts (rejects) a norm, but they erroneously assume most others reject (accept) it. To find evidence of pluralistic ignorance, we compared subjects' responses to NE and PNB questions. PNB questions had only yes/no answers and could not be incentivized.⁹

2.4. Procedure and pre-registration

The experiments were conducted on Amazon Mechanical Turk and programmed using the experimental platform Behav-ery.¹⁰ All subjects were from the US, 54.3% were male, and the average age was 34.2 (SD = 10.1).

We have pre-registered our experimental design and analyses before any data were collected (see Appendix Figure A1). The pre-registration document included our hypotheses, key dependent variables, all experimental conditions, estimated sample size, and exclusion criteria. We included two comprehension tests that verified subjects' understanding of the trust game, the specifics of their assigned condition and their compensation and punishment possibilities.¹¹ If a subject failed the test, they had to retake and pass it before continuing with the experiment. If the subject failed the tests more than twice,

⁷ A random-random (RR) condition where both players' decisions would be determined by a random machine was not conducted as no social norm or expectation of a player's action would be intentionally violated in that situation.

⁸ Previous research shows that the order of presentation of empirical/normative expectations does not affect behavior (Bicchieri & Xiao, 2009; Bicchieri & Chavez, 2010; Bicchieri et al., 2022).

⁹ We also elicited the empirical and normative expectations of observers about the actions of trust game players (Table A1 in Appendix). We found a complete symmetry about the expectations regarding the actions of trust game players and the expectations about other observers.

¹⁰ This software is tailored to interactive experiments where a number of subjects can make simultaneous decisions (<https://behavery.com/>)

Table 1

Elicitation of empirical expectations, normative expectations and personal normative beliefs of observers.

Panel A. No Reciprocity Outcome	
Empirical Expectations (EE)	<i>“What percentage of players in your role in this game have added money to Player A if Player A transferred the entire amount to Player B but did not receive any money back from Player B?”</i> <i>“What percentage of players in your role in this game have deducted money from Player B if Player A transferred the entire amount to Player B but did not receive any money back from Player B?”</i>
Personal Normative Beliefs (PNB)	<i>“Do you believe that a player in your role in this game should add money to Player A if Player A transferred their entire amount to Player B but did not receive any money back from Player B?”</i> <i>“Do you believe that a player in your role in this game should deduct money from Player B if Player A transferred their entire amount to Player B but did not receive any money back from Player B?”</i>
Normative Expectations (NE)	<i>“What percentage of players in your role in this game believe that one should add money to Player A if Player A transferred the entire amount to Player B but did not receive any money back from Player B?”</i> <i>“What percentage of players in your role in this game believe that one should deduct money from Player B if Player A transferred the entire amount to Player B but did not receive any money back from Player B?”</i>
Panel B. No Trust Outcome	
Empirical Expectations (EE)	<i>“What percentage of players in your role in this game have added money to Player B if Player B did not receive any money from Player A?”</i> <i>“What percentage of players in your role in this game have deducted money from Player A if Player B did not receive any money from Player A?”</i>
Personal Normative Beliefs (PNB)	<i>“Do you believe that a player in your role in this game should add money to Player B if Player B did not receive any money from Player A?”</i> <i>“Do you believe that a player in your role in this game should deduct money from Player A if Player B did not receive any money from Player A?”</i>
Normative Expectations (NE)	<i>“What percentage of players in your role in this game believe that one should add money to Player B if Player B did not receive any money from Player A?”</i> <i>“What percentage of players in your role in this game believe that one should deduct money from Player A if Player B did not receive any money from Player A?”</i>

Table 2

Experimental Design and Sample Sizes across Conditions.

Condition	Intentionality Investor	Intentionality Trustee	Number of ParticipantsNo Trust	Number of ParticipantsNo Reciprocity
Choice-Choice (CC)	Yes	Yes	75	83
Random-Choice (RC)	No	Yes	77	81
Choice-Random (CR)	Yes	No	65	82

they were later excluded from the dataset. This threshold was pre-registered and hence chosen before any data was analyzed to ensure that our MTurk subjects had an understanding of the experimental condition, available actions and consequences.

At the end of the experiment, subjects were asked to explain their compensation and punishment choices in an open-ended question and provided basic demographic information. Overall, a total of 545 subjects participated as observers, out of whom 82 were excluded from the analyses for failing the comprehension tests, leaving a total of 463 observers whose data were analyzed Table 2. below summarizes the experimental design and reports the sample sizes in all conditions. In the Appendix, we present the results on the full sample, without excluding the 82 observers who failed the comprehension tests. Their exclusion did not affect the results in any significant way. All players were anonymous. Sessions lasted around 10–15 min across conditions. The observers earned a \$1 participation fee and on average \$2.07 from their endowment after the adding and deducting choices they made.

2.5. Hypotheses

Let us summarize here our three main hypotheses.

H1: Reciprocity is a stronger social norm than trust.

H2: Third-party observers of trust games prefer to compensate the harm victims rather than punish the violators.

H3: Intentionality matters in deciding whether to compensate victims or punish norm violators.¹²

¹¹ We pre-specified several comprehension tests in our pre-registration due to recent concerns about the quality of responses on MTurk, high variance in MTurk subjects' attention and understanding of the basic experimental setup, as well as the presence of bots on MTurk platforms (Aruguete et al., 2019; Chmielewski & Kucker, 2020; Dennis et al., 2020). See Table A2 in Appendix for comprehension tests questions.

¹² Our pre-registration included an additional hypothesis on moral outrage and empathic concern as drivers of compensation and punishment decisions in intentional and random settings (Figure A1 in the Appendix). The questions aimed at measuring the level of moral outrage and empathic concern of our observers towards the trust game players and their situation (after seeing violations of trust or reciprocity) were the last questions included in the post-experimental questionnaire. As such, they did not have an influence on any subsequent actions or beliefs.

Table 3
Expectations and Beliefs of Observers (CC condition).

Action	Belief	No Trust	No Reciprocity	Difference between No Reciprocity and No Trust
Adding to Victims	Empirical Expectations	52.3 (3.8)	54.8 (3.1)	2.5 (4.8)
	Normative Expectations	55.7 (3.3)	63.0 (2.9)	7.3 (4.4)
	Personal Normative Beliefs	61.3 (5.7)	74.7 (4.8)	13.4* (7.4)
Deducting from Violators	Empirical Expectations	39.3 (3.4)	51.8 (3.1)	12.5*** (4.5)
	Normative Expectations	43.3 (3.4)	60.2 (2.9)	16.9*** (4.5)
	Personal Normative Beliefs	34.7 (5.5)	62.7 (5.3)	28.0*** (7.7)

Mean estimates by observers expressed in percentages (EE, NE) or percentages of observers (PNB). Wilcoxon-Mann-Whitney test used for testing differences. Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

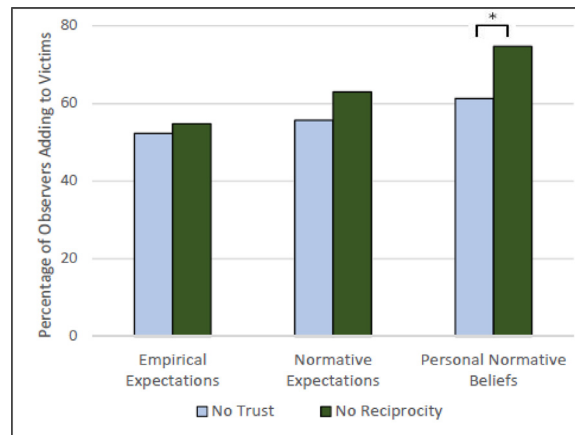


Fig. 2a. Expectations and Beliefs of Observers on Adding to Victims (CC condition).

3. Results

For the first two hypotheses that compare reciprocity and trust as social norms and the related compensation and punishment decisions of third-party observers, we keep the intentionality of decisions constant and analyze only the baseline CC condition. This condition is most similar to the standard trust game setup, where investors and trustees intentionally choose and implement their preferred course of action. The other two intentionality conditions (CR and RC) relate to our third hypothesis and they are examined in the intentionality analysis in [Sections 3.4 and 3.5](#).

3.1. Assessing expectations

For a rule of behavior to be a social norm, a first condition is that a majority of individuals must have empirical and normative expectations that most people conform to the behavior and approve of it, and favor punishing violators that deviate from this behavior ([Bicchieri, 2006](#)). We assess these expectations about other observers after they have made their decisions to punish/compensate the trust game players. To determine the relative strength of reciprocity and trust as social norms, we first examine the personal beliefs, as well as the empirical and normative expectations about punishing violators of the behavior. We then focus on how expectations about compensation differ when trust and reciprocity are violated.

The first section of [Table 3](#) shows the differences in expectations and personal beliefs about compensating victims when trust and reciprocity are violated. The observers believed that No-Trust and No-Reciprocity victims were equally compensated by other third parties (52.3% versus 54.8% of observers were expected to have added money to victims of trust and reciprocity violations, respectively, $p = 0.67$, WMW test) (empirical expectations). The observers also expected that other observers' approval of compensation of victims of lack of reciprocity would not be significantly different from the compensation of victims of lack of trust (63.0% versus 55.7%, respectively, $p = 0.10$, WMW test) (normative expectations). Similarly, 61.3% of observers believed they themselves should compensate victims of trust violation, while 74.7% believed they should compensate victims of reciprocity violation ($p = 0.07$, WMW test) (personal normative beliefs) (see [Fig. 2a](#)).

The second section of [Table 3](#) shows the differences in expectations and personal beliefs about deducting from violators when trust or reciprocity are violated (see also [Fig. 2b](#)). When investors exhibit a lack of trust, 39.3% of observers were expected to deduct money from non-trusting investors (empirical expectations). Observers expected that 43.3% of other observers approved of deducting money from non-trusting investors (normative expectations). In addition, only 34.7% of observers believed that they themselves should deduct money from non-trusting investors (personal normative beliefs).

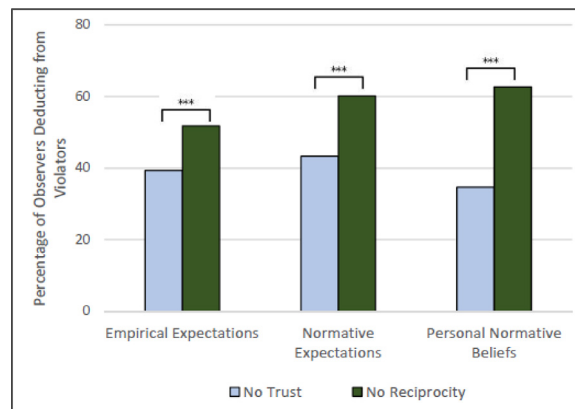


Fig. 2b. Expectations and Beliefs of Observers on Deducting from Violators (CC condition).

There was no majority of subjects that privately supported punishment for lack of trust, and they did not expect a majority of others to support punishment either.

On the other hand, when reciprocity is violated and trustees did not send half of the money back to the investors, observers estimated that 51.8% of players in their role deducted money from non-reciprocating trustees (empirical expectations). In terms of normative expectations, observers estimated that a majority of them (60.2%) approved deducting money from non-reciprocating trustees. In line with these results, most observers (62.7%) also believed that they themselves should deduct money from non-reciprocating trustees (personal normative beliefs). Since a majority of subjects privately supported punishment for the lack of reciprocity and expected most others to support it, we find strong evidence that reciprocity is a social norm and no evidence of pluralistic ignorance.

The last column of Table 3 shows the differences between the beliefs regarding adding and deducting when trust or reciprocity are violated.¹³ All three types of beliefs are significantly higher for deducting from violators in case of reciprocity violations, providing additional support to our first hypothesis and demonstrating that reciprocity is a strong social norm whereas trust is not. Observers tended to believe that not reciprocating is a worse violation than not trusting. Though one might expect to be trusted, disregarding such expectation does not rise to the level of a major transgression.

Taken together, the data support our hypothesis that reciprocity is a strong norm, trust is not, and also show that compensating victims whatever the cause of harm seems to be a social norm as well. Compensation did not significantly depend on the strength of the norm broken, or even if what was infringed was a norm. However, as we show in the next Section 3.2, higher compensation amounts were given to victims of a social norm (reciprocity) violation.

3.2. Norms in action

We are also interested in whether observers' empirical expectations and normative expectations are associated with their actions. For a social norm to exist, it is not sufficient that the necessary social expectations are present: a further condition is that individuals must also have a conditional preference to obey the rule, given their expectations (Bicchieri, 2006). In other words, social expectations need to drive behavior. Overall, around 60% of observers used at least one of these mechanisms (compensation or punishment), while the rest did not use any and decided not to change the trust game payoffs. Conditional on acting, almost all observers (95%) compensated the victims, significantly less (39%) punished the violators and 37% overall used both mechanisms.¹⁴

Table 4 shows the amounts and frequencies of victim compensation and punishment of violators in the No Trust and No Reciprocity conditions. The adding and deducting actions of our observers indicate that violating reciprocity invited stronger responses than violating trust, in line with their expectations. Both adding and deducting are significantly higher and more likely when reciprocity is violated. 43.4% of observers deducted from violators of reciprocity, while only 13.3% did so from non-trusting investors ($p < 0.001$, WMW test). The average amount deducted from violators of reciprocity was -£3.93, while observers deducted on average an amount insignificantly different from zero from non-trusting investors ($p < 0.001$, WMW test). Similarly, 44.0% of observers compensated No-Trust victims while 61.4% compensated No-Reciprocity victims ($p = 0.03$, WMW test). The average amount added to No-Trust victims was £2.35, while the amount given to No-Reciprocity victims was over three times higher, at £7.26 ($p < 0.001$, WMW test). For simplicity, we express these results through Figs. 3a and 3b that clearly show that violation of reciprocity is taken much more seriously than violation of trust.

¹³ Marginally lower values of empirical expectations compared to normative expectations and personal normative beliefs can be explained by the costs of adding and deducting actions that run counter to subjects' self-interest.

¹⁴ See Table A4 in the Appendix for details and summary statistics.

Table 4
Compensation of Victims and Punishment of Violators (CC condition).

Outcome Condition	Variable	Adding to Victims	Deducting from Violators	Difference between Adding and Deducting
No Trust ($N = 75$)	Frequency (%)	44.0 (5.8)	13.3 (4.0)	30.7*** (5.7)
	Amount (€)	2.35 (0.48)	0.91 (0.46)	3.25*** (0.84)
No Reciprocity ($N = 83$)	Frequency (%)	61.4 (5.4)	43.4 (5.5)	18.1*** (4.9)
	Amount (€)	7.26 (1.14)	−3.93 (0.73)	3.33* (1.39)

Standard errors in parentheses. McNemar's exact test used for differences in frequencies and Wilcoxon-Mann-Whitney test for differences in amounts.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

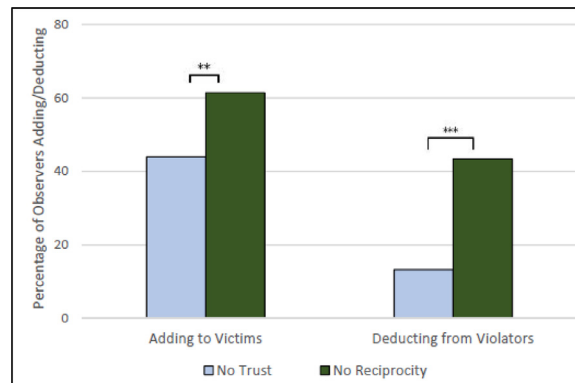


Fig. 3a. Frequency of Observers Compensating Victims and Punishing Violators (CC condition).

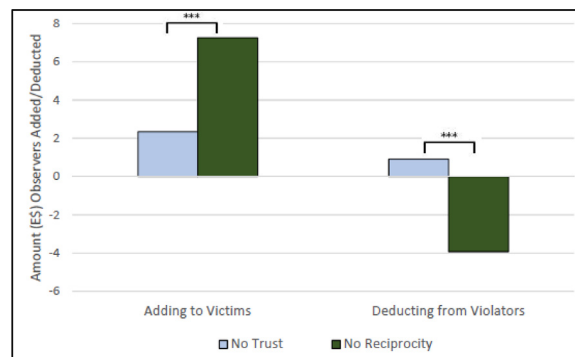


Fig. 3b. Amounts (€) of Observers' Compensation to Victims and Punishment of Violators (CC condition).

As mentioned previously, social norms are often supported by punishment and one way to diagnose the presence of a norm is to gauge the collective willingness to sanction its transgression. Our punishment results support the hypothesis that lack of reciprocity is a major transgression, whereas lack of trust is not equally critical. Though compensation does not signal the presence of a norm (or its violation), for some it may constitute a positive reinforcement of the norm, in addition to other motives such as empathy or inequality aversion.

In the correlation matrices below, we see a strong relationship between empirical and normative expectations in terms of both adding to victims and deducting from violators. The conditions of No Trust and No Reciprocity differ with respect to the strength of correlations between deducting actions and beliefs. In the No Reciprocity condition (Table 5a) subjects' empirical expectations, normative expectations and personal normative beliefs about deducting from violators are moderately to strongly correlated with their actual deducting actions.

In the No Trust condition (Table 5b) such beliefs are only weakly to moderately correlated with actions. These findings underline our result that reciprocity is a norm, while trusting is not. Consistent with our previous findings, we find no such differences between conditions when looking at the relationship between beliefs and actions regarding victims' compensations.

Table 5a

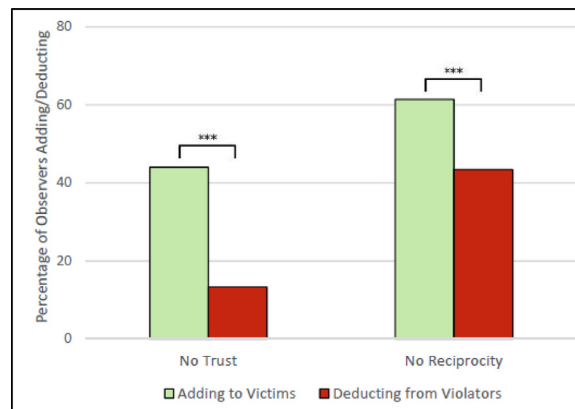
Correlation Matrix for Actions and Beliefs of Observers in No Reciprocity Conditions.

	Add	Deduct	EE Add	NE Add	PNB Add	EE Deduct	NE Deduct	PNB Deduct
Add	1							
Deduct	0.59***	1						
EE - Add	0.60***	0.53***	1					
NE - Add	0.41***	0.40***	0.61***	1				
PNB - Add	0.56***	0.34***	0.47***	0.50***	1			
EE - Deduct	0.43***	0.58***	0.45***	0.43***	0.26**	1		
NE - Deduct	0.48***	0.47***	0.46***	0.60***	0.26**	0.72***	1	
PNB - Deduct	0.31***	0.58***	0.25**	0.33***	0.41***	0.48***	0.52***	1

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.**Table 5b**

Correlation Matrix for Actions and Beliefs of Observers in No Trust Conditions.

	Add	Deduct	EE Add	NE Add	PNB Add	EE Deduct	NE Deduct	PNB Deduct
Add	1							
Deduct	0.36***	1						
EE - Add	0.62***	0.33***	1					
NE - Add	0.60***	0.30***	0.86***	1				
PNB - Add	0.65***	0.23**	0.44***	0.59***	1			
EE - Deduct	0.35***	0.38***	0.55***	0.49***	0.28**	1		
NE - Deduct	0.25**	0.25**	0.40***	0.51***	0.35***	0.68***	1	
PNB - Deduct	0.43***	0.37***	0.31***	0.45***	0.52***	0.39***	0.50***	1

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.**Fig. 4a.** Frequency of Observers Compensating Victims and Punishing Violators (CC condition).

3.3. Compensation versus punishment

Table 4 also showed the differences in compensation and punishment behavior within the No Trust and No Reciprocity conditions. Overall, we see that third parties prefer to compensate the victims rather than punish the violators, supporting our second hypothesis.

In the No Trust condition, about 44.0% of observers added to victims, but significantly fewer of them (13.3%) deducted from offenders ($p < 0.001$, McNemar's exact test) (Fig. 4a). Comparing amounts, we find that the average amount added to victims was E\$2.35, while the average amount deducted from offenders was a positive E\$0.91 ($p < 0.001$, Wilcoxon signed rank test) (Fig. 4b). In the No Reciprocity condition, 61.4% of observers added to victims compared to 43.4% who deducted from violators ($p < 0.001$, McNemar's exact test). The amount added to victims was E\$7.26 compared to -E\$3.93 deducted from violators, which is marginally significant ($p = 0.07$, Wilcoxon signed rank test), due to stronger preferences for punishing violators of reciprocity.

Our data show, once again, that punishment is strongly tied only to violations of a widely-held norm (reciprocity) and, while compensation seems to be indifferent to the cause of harm (no trust or no reciprocity), there is clearly an interaction between compensation and the violation of a strong norm. Compensation is higher when the victim is the trusting investor compared to a trustee who received no money ($p < 0.001$, WMW test).

What motivates observers to compensate victims and/or punish violators when it costs them money? Conditional on acting, significantly more observers have decreased the differences in final payoffs between investors and trustees (62%)

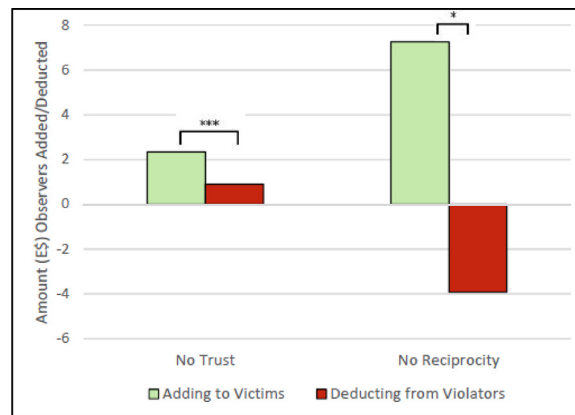


Fig. 4b. Amounts (€) of Observers' Compensation to Victims and Punishment of Violators (CC condition).

Table 6

Actions of Observers across Intentionality Conditions.

Action	Condition	No Trust		No Reciprocity	
		Frequency (%)	Amount (€)	Frequency (%)	Amount (€)
Adding to Victims	CC	44.0 (5.8)	2.35 (0.48)	61.4 (5.4)	7.26 (1.14)
	RC	55.8 (5.7)	2.66 (0.46)	58.0 (5.5)	5.40 (0.97)
	CR	53.8 (6.2)	2.47 (0.44)	64.6 (5.3)	7.02 (1.05)
Deducting from Violators	CC	13.3 (4.0)	0.91 (0.46)	43.4*** (5.5)	-3.93*** (0.73)
	RC	13.0 (3.9)	1.16 (0.55)	35.8** (5.4)	-2.52*** (0.57)
	CR	13.8 (4.3)	0.66 (0.51)	17.1***, ** (4.2)	0.20***, *** (0.59)

Standard errors in parentheses. Comparisons made across intentionality conditions. Dunn's test using the Sidak adjustment for multiple comparisons.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

compared to observers that have increased the differences (31%) ($p = 0.002$, Wilcoxon signed rank test) or kept them the same (7%) ($p < 0.001$, Wilcoxon signed rank test).¹⁵

3.4. Intentionality and actions

To examine the effects of intentionality on the beliefs and actions of observers, we expand our analyses to the Choice-Random (CR) and Random-Choice (RC) conditions and compare them to the baseline Choice-Choice (CC) condition. As explained in the Experimental Design section, in the CR condition the investor's action is voluntary, but the trustee's action is not, as a random machine decides for her and has an 80% chance of sending the money back to the investor. Similarly, in the RC condition, the investor's action is involuntary but the trustee's is voluntary. The random machine makes the decision for the investor and sends the money to the trustee with a 50% probability. These conditions were common knowledge among the players and the observers.

Table 6 examines how intentionality interacts with observers' actions when a strong norm of reciprocity is broken compared to when trust is lacking. There are no significant differences across the intentionality conditions in terms of compensating victims, regardless of whether trust or reciprocity were violated. The observers did not discriminate between the amounts they added to the victims based on the intentionality of harmful actions (Figure 5b1), nor were the frequencies of compensation significantly different (Figure 5a1).

On the other hand, the punishment of violators varies with the intentionality of players' actions, but only when a social norm (reciprocity) is violated. There were no significant differences in punishment across the intentionality conditions when trust was absent, either for deciding to punish at all, or in terms of average punishment amounts. When reciprocity was violated, however, the punishment frequency (17.1%) and amount (€0.20) in the CR condition were significantly lower

¹⁵ At the end of our experiment, observers filled out a required text box explaining their compensation and punishment decisions. We coded their responses according to all the motives they mentioned. The act of compensating the victims was strongly negatively correlated with self-interest ($r = -0.78$), moderately positively correlated with fairness or equality concerns ($r = 0.53$), empathy ($r = 0.33$), negative emotions towards the violator ($r = 0.34$), and weakly positively correlated with altruism towards both players ($r = 0.20$). The act of deducting from violators was moderately negatively correlated with self-interest ($r = -0.51$), moderately positively correlated with negative emotions towards the violator ($r = 0.51$), fairness or equality concerns ($r = 0.32$), and weakly positively correlated with their concerns about morality ($r = 0.22$), restoring the norm ($r = 0.24$) and empathy towards the victim ($r = 0.19$). As can be seen in Appendix (Table A9), a majority of observers said they were motivated by fairness, equality or both.

Table 7
Expectations and Beliefs of Observers across Intentionality Conditions.

Action	Outcome Condition		CC	RC	CR
Adding to Victims	No Trust	Empirical Expectations	52.3 (3.8)	57.1 (3.0)	50.4 (3.7)
		Normative Expectations	55.7 (3.3)	53.2 (2.9)	52.7 (3.4)
		Personal Normative Beliefs	61.3 (5.7)	70.1 (5.3)	61.5 (6.1)
	No Reciprocity	Empirical Expectations	54.8 (3.1)	50.6 (3.1)	56.1 (3.6)
		Normative Expectations	63.0 (2.9)	54.6 (3.3)	58.8 (3.5)
		Personal Normative Beliefs	74.7 (4.8)	71.6 (5.0)	65.9 (5.3)
Deducting from Violators	No Trust	Empirical Expectations	39.3 (3.4)	34.4 (2.8)	35.4 (2.6)
		Normative Expectations	43.3* (3.4)	33.8*,** (2.9)	44.2** (2.9)
		Personal Normative Beliefs	34.7 (5.5)	31.2 (5.3)	30.8 (5.8)
	No Reciprocity	Empirical Expectations	51.8* (3.1)	46.3 (3.0)	43.3* (3.3)
		Normative Expectations	60.2*** (2.9)	53.4 (3.1)	45.7*** (3.3)
		Personal Normative Beliefs	62.7*** (5.3)	63.0*** (5.4)	30.5***,*** (5.1)

Notes: Mean estimates by observers expressed in percentages (EE, NE) or percentages of observers (PNB). Standard errors in parentheses. Comparisons are made across intentionality conditions. Dunn's test using the Sidak adjustment for multiple comparisons. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

compared to punishment in the CC condition (at 43.4% and of -E\$3.93) and RC condition (at 35.8% and of -E\$2.52) (Figures 5a2 and 5b2).¹⁶ Therefore, the level of punishment was substantially reduced if a social norm is broken involuntarily, as in the CR condition a random machine made the decision to not reciprocate. In contrast, in both the CC and RC conditions, the trustee's action of not reciprocating was voluntary. We can conclude that intentionality does matter, but only when deciding whether and how much to punish the violators of social norms.¹⁷

3.5. Intentionality and beliefs

Lastly, we examine if intentionality affected the personal beliefs and social expectations of observers about compensating victims and punishing violators Table 7. displays these beliefs/expectations across the three intentionality conditions with respect to trust and reciprocity violations. Similar to Table 3, we see no significant differences across empirical, normative expectations and personal normative beliefs when it comes to compensating victims in both trust and reciprocity violations. The observers believed that victims have been and should be compensated similarly regardless of the intentions behind the lack of trust or reciprocity. Intentions instead had a major effect on the beliefs and expectations regarding the punishment of violators. Yet, as we see below, not all intentional violations are equal. Intentional violations of trust are more easily condoned than intentional violations of reciprocity.

With respect to offenders of trust, observers expected the lowest percentage of other observers to believe that they should deduct money from the investor when the violation was unintentional (33.8% in the RC condition) compared to intentional violations of trust by investors in the CC (43.3%) ($p = 0.06$, Dunn's test, Sidak adjustment) and CR (44.2%) conditions ($p = 0.03$, Dunn's test, Sidak adjustment). There was no significant difference with respect to personal normative beliefs or empirical expectations across intentionality conditions as observers estimated a similar percentage of other observers to have punished both the intentional and unintentional offenders of trust.

We again see a more notable effect of intentions when deciding to punish the violators of reciprocity. With normative expectations, observers estimated that 60.2% of other observers approved of punishing trustees for intentional lack of reciprocity after an intentional show of trust (CC condition), whereas 53.4% of observers were believed to approve of punishment when trust was unintentional (RC condition) ($p = 0.21$, Dunn's test, Sidak adjustment). If the reciprocity violation was unintentional (CR condition), significantly fewer observers (45.7%) were believed to approve the punishment for lack of reciprocity compared to when the violation was intentional in CC condition ($p = 0.003$, Dunn's test, Sidak adjustment). Personal normative beliefs show a similar pattern. Significantly more observers in the CC (62.7%) and RC (63.0%) conditions believed that one should punish the trustees that intentionally did not reciprocate in comparison to unintentional violations of reciprocity by a random machine in the CR condition (30.5%) ($p < 0.001$, Dunn's test, Sidak adjustment). Regarding empirical expectations, there was only a marginal increase in the estimated percentage of observers that have punished intentional violations of reciprocity (51.8%, CC condition) compared to unintentional violations by a random machine (43.3%, CR condition) ($p = 0.09$, Dunn's test, Sidak adjustment).^{18,19}

¹⁶ We find significant differences between CC and CR conditions ($p < 0.001$, Dunn's test, Sidak adjustment for probabilities; $p < 0.001$, Dunn's test, Sidak adjustment for the amounts) and between CR and RC conditions ($p = 0.02$, Dunn's test, Sidak adjustment for probabilities; $p < 0.001$, Dunn's test, Sidak adjustment for the amounts). The direction and significance of these results remain the same without exclusions of subjects that failed the comprehension tests (see Appendix Table A10).

¹⁷ Frequency of deducting from unintentional offenders of reciprocity in the CR condition (17.1%) was closer to the frequency of deducting from offenders of trust across all intentionality conditions (13.0%–13.8%) than to the frequency of deducting from intentional violators of reciprocity (CC and RC conditions).

¹⁸ Marginal differences in the empirical expectations can be explained by the costs associated with adding and deducting actions that run counter to the observers' self-interest.

In both No-Trust and No-Reciprocity outcomes in the CR condition, we observe that normative expectations about punishment are significantly higher than personal normative beliefs (44.2% vs. 30.8% in No Trust, 45.7% vs. 30.5% in No Reciprocity). Lower levels of personal normative beliefs are quite rational because they show that the observers understand that lack of intentionality (randomness in the responders' action) would justify the investor not trusting and would also justify much less punishment for the non-reciprocators. The differences between normative expectations and personal normative beliefs seem to indicate that observers may have doubts about other observers' understanding of the consequences of the lack of intentionality of trustees' actions. It is also noticeable that a minority of 33.8% of observers think others believe one should punish lack of trust in the RC condition, where the investor had no choice. If we look at observers' underlying motivations as stated in a follow-up survey (Appendix Table A9), it appears that fairness and equalization of outcomes are significant motivators. It is possible that this normative expectation might be due to observers' beliefs that at least some other observers aimed to equalize outcomes.

4. Discussion

When people witness norm violations, they can reinforce those norms and modify the final payoffs of the parties by using compensation and punishment. In line with prior research, we find that third-party observers prefer to compensate the victims rather than punish the violators, supporting our second hypothesis (Chavez and Bicchieri, 2013; Leliveld et al., 2012; Lotz et al., 2011; Van Doorn et al., 2018). How do intentions affect this preference? Our third hypothesis was partially supported by the data as intentions mattered when deciding if and how much to punish, but did not affect compensation decisions. Subjects did and were expected to compensate for harm regardless of whether a norm was broken or the intentions of the harming party. The focus of compensating behavior was the victim's unfavorable situation due to lack of trust or lack of reciprocity, not the intent behind the violations. There was uniform agreement that victims need to be compensated for both intentional and unintentional harm; the fact that the investor/trustee action was intentional or instead produced by a random machine did not matter when considering compensation. Though compensation per se was not signaling a norm violation, it appears to be a social norm itself in the presence of harm or unfair outcomes, as redressing harm was a widely expected and approved behavior by observers.

While compensation is a universal response to harm irrespective of its cause, punishment is highly selective. Our data show two relevant dimensions of punishment decisions. Punishment is conditional on the strength of the violated social norm, as well as the intent behind the norm violation. Subjects punish more frequently and more severely when strong norms are violated. However, if the norm violation was clearly unintentional, essentially no punishment occurred.

Our experimental design differs from previous trust games by being able to identify the existence and strength of social norms. First, we elicited beliefs about the behavior and beliefs of other observers (empirical and normative expectations). Strong and mutually consistent expectations suggest the presence of a social norm (Bicchieri, 2006). The majority of subjects supported and expected punishments of reciprocity violations, while only a minority did so for lack of trust. Their responses thus indicate that lack of reciprocity is much more blameworthy than lack of trust (Bicchieri et al., 2011). Though mutually consistent expectations suggest the presence of a norm, it is also important to assess whether behavior is conditional on such expectations.²⁰ Second, we show that subjects' actions and beliefs were strongly correlated, adding validity to our design and consistency to their behavior. Finally, both the actions and expectations of our subjects reveal that reciprocity is a social norm, while trust is not, lending support to our first hypothesis.

What about compensation? When asked about compensation, most participants expected victims to be compensated in all conditions, irrespective of the source of the violation (lack of trust or lack of reciprocity), and also expected others to support this behavior. Notably, the strength and consistency of compensation expectations suggested the presence of a norm, and indeed we found that subjects' actions and expectations were strongly correlated. In sum, participants' responses and behavior indicate that both reciprocity and harm compensation are strong social norms.

We also found that compensation is generally preferred to punishment across conditions. From an evolutionary viewpoint, it would seem that punishing transgressors not only strengthens social norms, but can also result from a deterrence psychology for defending personal interests, as mistreatment of a third party may predict mistreatment of oneself later on (Delton and Krasnow, 2017; Krasnow et al., 2016), and even serve as a reputation enhancer (Jordan et al., 2016; Jordan and Rand, 2020). So third-party punishment, even when quite costly, can be motivated by self-interest and reputation management, though punishment may also have drawbacks. Studies show, for example, that punishment that is not clearly legitimized and justified may invite retaliation (Bicchieri et al., 2021; Xiao, 2013), and punishers may be even willing to pay to hide their punitive behavior (Rockenbach and Milinski, 2011).

Compensation *prima facie* seems to have few advantages. For one, it could be exploited by transgressors, thus weakening the fabric of society. Compensation does not restrain transgressions and may even make transgressions appear less damaging as there will be some sort of harm reparations. On the other hand, compensation can be seen as a form of insurance against all sorts of harms. The very existence of a society is predicated on offering its members some security against

¹⁹ The direction and significance of these results remain the same without exclusions of subjects that failed the comprehension tests (see Appendix Table A11).

²⁰ Conditional preferences are a marker for social norms, whereas strong and mutually consistent expectations may also indicate shared moral rules, where actions are not interdependent. For a moral rule, what others do/approve of is not a reason for action (Bicchieri, 2006).

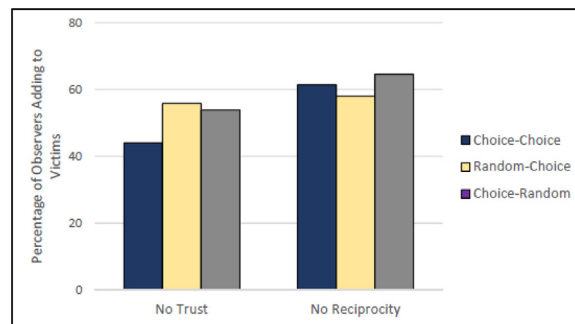


Fig. 5a1. Percentage of Observers Adding to Victims across Intentionality Conditions.

adverse events. In this sense, a society without some form of harm compensation would not last long. Indeed, the preference for compensation has deep historical roots. There is ample evidence that early acephalous (non-state) societies strongly favored restitution/compensation over retaliation/punishment (Michalowski, 1985).²¹ In fact, while restitution often takes something away from the offender, it is different in form, purpose and consequences from punishment as retribution or revenge (Weitekamp, 1993).²²

From a social welfare viewpoint, punishment corresponds to a net loss, as it decreases both the payoff of the punisher and offender. Compensation, on the other hand, balances the decrease in the compensator's payoff with an increase in the victim's payoff, so it is a Pareto superior outcome to punishment. If we consider also reputational effects, studies show that when third parties are seen as free to choose between compensation, punishment, and doing nothing, compensators are preferred to punishers and non-interveners as game partners (Ozono and Watabe, 2012). So compensation, too, can win reputational benefits, as the concern for fairness is accompanied by a show of kindness and empathy (Gordon et al., 2014). Beyond reputation management, the choice to compensate may also be linked to the preference for avoiding harming others. For example, in a recent study Hart et al. (2019) show that observers, when given a choice, prefer to reveal random harm to victims (to ease their suffering) rather than intentional harm.

Some studies also show that different causes of harm elicit different emotions. As we have seen, harm can be related or unrelated to a norm violation. For example, Thulin and Bicchieri (2016) show that moral anger is the dominant emotion when compensation is linked to a norm violation, whereas empathy is the main emotion present when we compensate for generic harms.²³ Moral anger is mainly related to wrongful deeds, whereas empathy is associated with harmful consequences and it might be argued that the former is a stronger, more focused emotion than the latter (Hechler and Kessler, 2018). Though we observed an overall preference for compensation, the amount of compensation in our Choice-Choice (CC) condition was more frequent and much higher when reciprocity was violated rather than trust (Figs. 3a and 3b).

Finally, the added element of intentionality behind norm violations in our design proved crucial for punishment decisions and expectations. Punishment mainly occurs when the norm violation is intentional. The focus of the decision-making process is now the violator and the nature of their norm-breaking action. Unintentional violations of reciprocity invited virtually no punishment as did intentional and unintentional violations of trust. Note that unintentional violations do not signal that a norm (reciprocity) is not present, only that its violation is not punishable due to lack of intent and hence responsibility. However, intentional violations of reciprocity, even when coupled with unintentional trust, were punished by third parties. Compensation, being only associated with reparation of harmful consequences, was not impacted by intentions or norm strength. Though, as we have discussed, compensation was more frequent and generous in the presence of a norm violation, it was still preferred to punishment across all conditions. There were no differences in personal beliefs and expectations (EE, NE, PNB) with respect to compensation as intentions only affected punishment of transgressors.

In decreasing payoff differences between players or remedying damages due to norm violations, people tend to follow a pecking order of first compensating the victims and then punishing the violators in case a social norm has been violated with intent. Punishment is fine-tuned to norm violation *and* intentionality, compensation is more generic and gives similar weights to the causes of harm. Punishment and compensation hence have unequal signaling power when it comes to detecting the presence of social norms. When we observe punishment, we can conclude that a norm has been violated. When we see compensation, the signal is not as clear.

Our findings contribute to explanations of the conditions under which selective and higher-order punishment of non-punishing observers would occur (Martin et al., 2019). It also contributes to the literature on the differences between third-party and second-party punishments (Kriss et al., 2016), and the importance of signaling intentions for trust-reciprocity

²¹ Nader and Combs-Schilling (1977, pp. 34–35) discuss the purposes and functions of the restitution process in acephalous societies.

²² Reporative justice was also common in early state societies, and only by the end of the twelfth century did we see the erosion of reparative justice (Schafer, 1975).

²³ See also Akerlof (2016) and Hart et al. (2019).

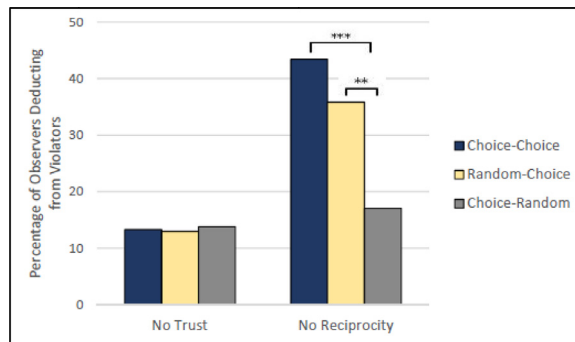


Fig. 5a2. Percentage of Observers Deducting from Violators across Intentionality Conditions.

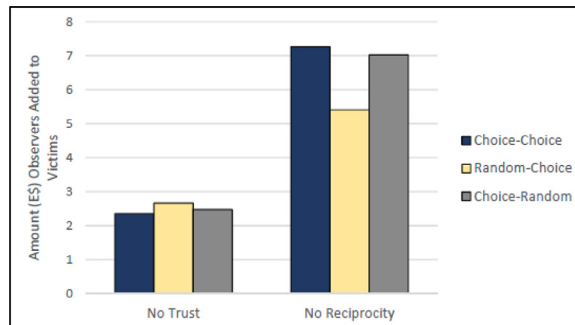


Fig. 5b1. Amounts Observers Added to Victims across Intentionality Conditions.

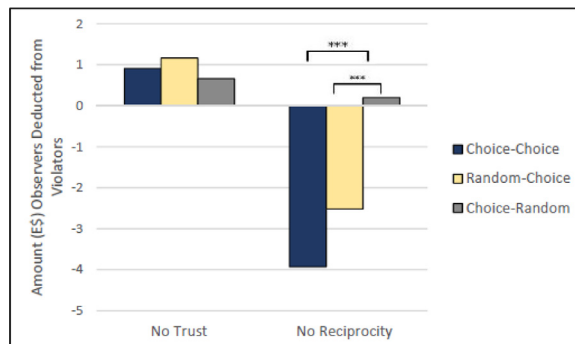


Fig. 5b2. Amounts Observers Deducted from Violators across Intentionality Conditions.

outcomes or contributions to public goods (Kessler, 2017; Toussaert, 2017). Further research is needed to understand the psychological and neurocognitive mechanisms underlying punishment and compensation decisions in the face of norm violations (Civai et al., 2019; Stallen et al., 2018).

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Declarations of Interest Statement

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jebo.2022.02.026](https://doi.org/10.1016/j.jebo.2022.02.026).

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